Track 4

Title: Towards translational and mechanistic whole-brain simulations with The Virtual Brain

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About:

In this workshop, participants will learn the basic concepts and methods for personalized whole-brain network modeling using the open-source neuroinformatics platform **The Virtual Brain** (TVB; thevirtualbrain.org). Participants will learn how to create individualized whole-brain models, run simulations, and use supporting neuroinformatics tools, including the latest extensions to The Virtual Brain Ontology and the optimization tool TVB-Optim.

The Virtual Brain Ontology (TVB-O, virtual-twin.github.io/tvbo) provides a knowledge framework for brain network modeling, including FAIR model specification and automated code generation, as well as a bridge to biomedical ontologies. It helps convert published models or one's own computational research questions into executable, reproducible simulation experiments and provides tools for analysis and visualization.

TVB-Optim (virtual-twin.github.io/tvboptim) brings gradient-based optimization to TVB, enabling the tuning of up to thousands of parameters simultaneously, allowing users to investigate brain dynamics at an unprecedented scale. Seamless parallelization and scaling to GPUs allow additional performance improvements, compared to base TVB, that are orders of magnitude.

Format: In a Hands-On session, participants will be introduced to the theoretical and programmatic framework of specifying, running, analyzing, and optimizing dynamical brain network models.

Agenda:

Day 1:

1. Intro to The Virtual Brain and its Applications
    - Basic Ideas and History
    - Mathematical Framework
    - Applications in Basic Research
    - Clinical Applications
2. Defining a FAIR Brain Network Model
    - Mean Field Models (+ Hands on)
    - Network of coupled dynamics (+ Hands on)
3. Dynamical Perspective: How to analyze a brain system
    - How to perform and interpret Bifurcation Analyses
4. Parameter Exploration & Model Optimization
    - Brain Network Models as a computational hypothesis of the brain.
    - Use-Case A: Mechanistic simulation of Alzheimer's Disease. Learning how to integrate biomedical data into brain network models.
    - Use-Case B: Homogeneous vs. Heterogeneous Parameter Fitting of Frequency Gradients (or any other map from TVBase).
5. Stimulating the Brain
    - (Visual/TMS) Evoked Potentials (JR)

Day 2 - Mini Hackathon:

We help you to answer and gain insight into your own research questions. Feel free to bring your data.

Target Audience:

Researchers with an interest in Computational Neuroscience, including data scientists, clinicians, engineers, and, in general, the curious. Prior experience with TVB is helpful but not mandatory.

Requirements:

- Basic Python programming skills are recommended.
- Notebooks can be run online (EBRAINS account required). Limited support for local setups will also be provided.

References: